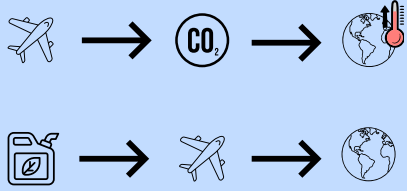


Multi-Objective Optimization of Sustainable Aviation Fuel (SAF) Blends Using a Physics- and Kinetics-Informed Turbofan Digital Twin

Introduction



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Aviation accounts for approximately 2.5% of global CO₂ emissions. SAFs can reduce these emissions by 80% compared to jet fuel.

Experimental blend testing is expensive, time-intensive, and limited.

Existing tools such as Computational Fluid Dynamic (CFD) simulations model individual configurations well but cannot efficiently search a high-dimensional blend space. (Wang et al., 2024)

Prior Work

Nath et al. (2023): PINNs predict gas flow dynamics and unknown parameters in diesel engines with limited, noisy data.

Ma et al. (2026): Locally-enhanced PINNs achieve <1% average wall-temperature error in aero-engine nozzle modeling.

Kuzhagaliyeva et al. (2022): AI + genetic algorithms design fuel mixtures from molecular descriptors, reducing lab testing cost.

Gal et al. (2024): 0D/1D Cantera + 3D CFD for hydrogen engine; holistic modeling essential for accuracy.

Purpose

What blend of sustainable aviation fuels can be identified through hybrid Cantera and physics-informed neural network simulations to minimize emissions while preserving jet engine performance under realistic thermodynamic conditions?

Methodology

Fuel Blend Modeling

HEFA-SPK, FT-SPK, ATJ-SPK + Jet-A modeled using CRECK C1-C16 Surrogates

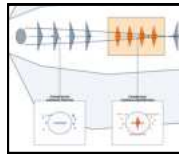


Figure 1. Cantera Architecture
Made by researcher with Matplotlib in Python

Turbine and Nozzle PINN Modeling

Fuel-dependent neural surrogate; 8D fuel-aware input, physics gates active, BC err <0.5%; exact continuity enforced

Emissions Modeling

NO_x: ICAO reg.; R²=0.9969; CO₂: LCA×3.16 kg/kg

Cantera Component Modeling

Brayton cycle compression work; 1700+ reactions.
OPR → T, p, p, cp, R, γ

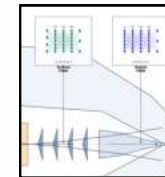


Figure 2. PINN Architecture
Made by researcher with Matplotlib in Python

Optuna Optimizer

1,000 trials; 4 objectives (TSFC, NO_x, CO₂, Thrust)

Results

4-objective optimization (minimize TSFC, CO₂, NO_x; maximize specific thrust) over a continuous SAF blend space (HEFA, FT-SPK, ATJ-SPK fractions + Jet-A baseline). Pareto filtering retained 398 of 1,000 trials as non-dominated solutions. Specific thrust remained stable (800.5–805.8 N/s/kg, ±0.7%) across all 398 blends, confirming performance preservation.

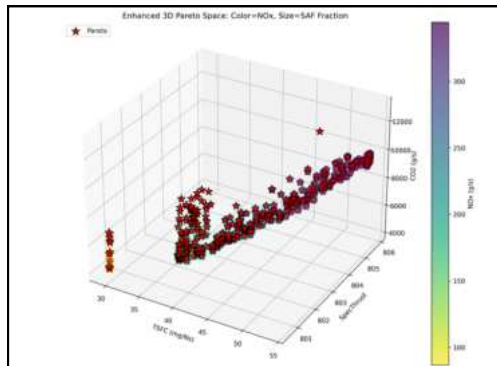


Figure 3. 3D Pareto Objective Space (TSFC · CO₂ · Specific Thrust)
Made by researcher with Matplotlib in Python

Each star is a Pareto-optimal blend: no improvement on one objective without worsening another.

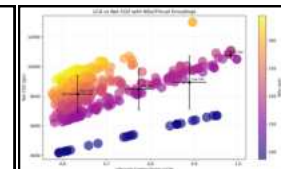


Figure 4. Lifecycle Carbon Factor vs. Net CO₂ Emissions
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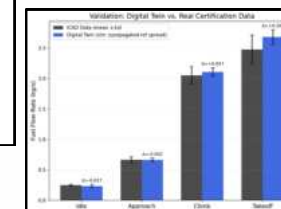


Figure 5. ICAO LTO Fuel Flow Validation (Rolls-Royce Trent 1000)
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Conclusion

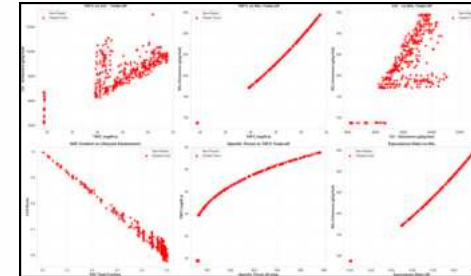


Figure 6. 2D Pareto Trade-off Panels
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Data-based methods such as PINNs and Cantera can be used to screen SAFs

398 Pareto-optimal blends identified from 1,000 simulated engine cycles

Best CO₂ blend (~49% SAF: HEFA + FT-SPK + ATJ-SPK) achieves ~51% lifecycle CO₂ reduction vs. Jet-A while maintaining specific thrust within ±0.7%

71% of Pareto solutions contain >40% SAF, confirming high-SAF blends dominate the optimal performance-emissions region; fuel efficiency (TSFC) and emissions reduction are co-aligned objectives

Future Work

Experimental combustion testing of top Pareto blends (e.g. lab calorimetry)

Extend to 2D planar nozzle (LE-PINN) once training instability is resolved

Fleet-level lifecycle mission simulation with full altitude/payload/range profiles.

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